

References

1. Etzkorn JR, Sobanko JF, Elenitsas R, Newman JG, et al. Low recurrence rates for in situ and invasive melanomas using Mohs micrographic surgery with melanoma antigen recognized by T cells 1 (MART-1) immunostaining: tissue processing methodology to optimize pathologic staging and margin assessment. *J Am Acad Dermatol* 2015;72:840–50.
2. Cheraghlu S, Christensen SR, Agogo GO, Girardi M. Comparison of survival after Mohs micrographic surgery vs wide margin excision for early-stage invasive melanoma. *JAMA Dermatol* 2019;155:1252–9.
3. Xiong DD, Barriera-Silvestrini P, Knackstedt TJ. Delays in the surgical treatment of melanoma are associated with worsened overall and melanoma-specific mortality: a population-based analysis. *J Am Acad Dermatol* 2022;87:807–14.
4. Cheraghlu S, Agogo GO, Girardi M. System-level variations in treatment delay for nonmetastatic melanoma. *J Am Acad Dermatol* 2019;81:1399–401.
5. Baumann BC, MacArthur KM, Brewer JD, Mendenhall WM, et al. Management of primary skin cancer during a pandemic: multidisciplinary recommendations. *Cancer* 2020;126:3900–6.

Shayan Cheraghlu, MD*

Maressa C. Criscito, MD*

Mary L. Stevenson, MD*

John A. Carucci, MD, PhD*

*All authors are affiliated with the Ronald O. Perleman
Department of Dermatology, New York University Grossman
School of Medicine, New York, New York

Supplemental digital content is available for this article.
Direct URL citations appear in the printed text and are
provided in the HTML and PDF versions of this article on the
journal's Web site (www.dermatologicsurgery.org).

The authors have indicated no significant interest with
commercial supporters.

Deemed not to be human subjects research and therefore
exempt from IRB review by the NYU Langone IRB.

Retrospective Cohort Analysis of Sex Differences in American Indians and Alaska Natives With Invasive Melanoma From the National Cancer Database

The Indian Health Services provides care to 2.56 million of the approximately 5.2 million members of federally recognized American Indian and Alaska Native tribes in the United States.¹ Only 1% of American Indian and Alaska Native homelands have the recommended number of dermatologists to adequately care for the population,² limiting access to skin cancer screening and surveillance which can perpetuate health disparities in American Indian and Alaska Native individuals. Compared with “White” individuals (as defined by the National Cancer Database [NCDB] and other United States–based cancer registries including the Surveillance, Epidemiology, and End Results [SEER] registry, hereafter referred to as “fairer skinned”), American Indian and Alaska Native individuals with melanoma are more likely to present with later stages and they have worse melanoma-specific survival.³ In addition to race, sex also strongly influences melanoma outcomes.^{4,5} Women have different tumor characteristics and better survival compared with men in the general population.^{4,5} Owing to the rarity of melanoma in American Indian and Alaska Native individuals, little is known about how sex affects this population's presentation and outcomes. This study explores sex differences in tumor characteristics and survival between American Indian and Alaska Native men and women with invasive melanoma.

Methods

The NCDB 2004 to 2018 was queried for biopsy-confirmed cases of primary cutaneous invasive melanoma (Histology Codes 8,720/3–8,780/3; Primary Site Codes C44.0–C44.9) in American Indian and Alaska Natives without prior malignancies. Race in the NCDB is self-reported by patients. Melanoma in situ tumors were excluded due to the negligible impact on overall survival to facilitate a survival analysis. Statistical tests

included Mann–Whitney *U* (continuous variables), chi-squared (categorical variables), Kaplan–Meier with log-rank test (overall survival), and a multivariate extended Cox regression (independent overall survival predictors). *p* values <.05 were considered statistically significant.

Results

Of the 429 cases, 48.7% were males and 51.3% were females (Table 1). The median age at diagnosis was higher for men (57 years, interquartile range [IQR] 47–67) than women (52 years, IQR 40–62) (*p* < .001). There was a significant association between sex and primary site (*p* < .001), with higher rates on the trunk in men compared with women (38.3% vs 22.3%, respectively) and higher rates on lower extremities in women compared with men (32.7% vs 12.0%, respectively). Although the stage was not significantly associated with sex (*p* = .189), women had higher rates than men of Stage I disease (58.3% vs 47.6%) and lower rates of Stages II (16.2% vs 22.2%), III (18.1% vs 21.7%), and IV (7.4% vs 8.5%). Sex was associated with histologic subtype (*p* = .045); nodular melanoma was more common in men (12.9% vs 7.7%); acral lentiginous melanoma was more common in women (7.7% vs 2.9%). There was no significant difference in the proportion of men and women who underwent surgery, radiation, chemotherapy, or immunotherapy, nor was there a difference in Charlson–Deyo Comorbidity score (a score to represent the burden of comorbidities in patients).

The 5-year overall survival rates were higher for women than men (81.5% vs 68.1%; *p* = .008). After adjusting a Cox regression for confounders (age at diagnosis, primary site, insurance status, Charlson–Deyo Comorbidity score, stage at diagnosis, histologic subtype, ulceration, surgery, chemotherapy and/or immunotherapy, and radiation), male

TABLE 1. Baseline Demographics of Males and Females

Total <i>n</i> = 429	Males (<i>n</i> = 209; 48.7%)	Females (<i>n</i> = 220; 51.3%)	<i>p</i> *
Age at diagnosis (yr), median (IQR)	57 (47–67)	52 (40–62)	<.001
Breslow depth (mm), median (IQR)	1.20 (0.44–3.00)	0.91 (0.41–2.10)	.144
Stage at diagnosis†,‡	<i>n</i> (%)	<i>n</i> (%)	.189
I	90 (47.6)	119 (58.3)	
II	42 (22.2)	33 (16.2)	
III	41 (21.7)	37 (18.1)	
IV	16 (8.5)	15 (7.4)	
Ulceration present†	47 (24.9)	44 (22.2)	.540§
Primary site			<.001
Head and neck	47 (22.5)	32 (14.5)	
Trunk	80 (38.3)	49 (22.3)	
Upper extremities	39 (18.7)	55 (25)	
Lower extremities	25 (12)	72 (32.7)	
NOS/overlapping	18 (8.6)	12 (5.5)	
Melanoma histology			.045
NOS/other	119 (56.9)	119 (54.1)	
Superficial spreading	57 (27.3)	67 (30.5)	
Nodular	27 (12.9)	17 (7.7)	
Acral lentiginous	6 (2.9)	17 (7.7)	

Significant *p* values (<0.05) are in bold.

* *p* values were calculated using the Pearson chi-squared test for categorical variables. The Mann–Whitney *U* test was used for continuous variables.

† Missing values are present for the following variables (number of missing values): stage at diagnosis (36) and ulceration status (42). Missing values are not included in percentage calculations.

‡ The American Joint Committee on Cancer (AJCC) editions 6, 7, and 8 were used for staging, based on the year of diagnosis. When pathologic staging was not available, clinical staging was used. Stages were equated across all editions of the AJCC.

§ In comparison with negative findings.

IQR, interquartile range; NOS, not otherwise specified.

sex independently increased mortality risk compared with female sex (adjusted hazard ratio 1.67; 95% confidence interval 1.37–2.03; *p* < .001).

Discussion/Conclusion

Unlike “fairer skinned” individuals in which men are more commonly affected by melanoma and “Black” individuals (as defined by NCDB and SEER, considered here to be “darker skinned”) in which women are more commonly affected, our cohort had similar proportions of men and women, consistent with previous research in American Indian and Alaska Natives.³ The median age at diagnosis is older for men than women (61 vs 55 years) in the general population of primarily “fairer skinned” individuals,⁴ and American Indian and Alaska Native men were also diagnosed later than women (57 vs 52 years), but American Indian and Alaska Native patients of both sexes were diagnosed 4 to 5 years earlier than the general population. This could be due to earlier onset of melanoma or more aggressive tumor behavior in American Indian and Alaska Natives compared with the general population, rather than earlier detection, which is less likely given limited access to dermatologic care among American Indian and Alaska Native patients,² and later stages at diagnosis in American Indian and Alaska Native than “fairer skinned” patients.³

The primary site distribution by sex in American Indian and Alaska Natives corroborates previous research in predominantly “fairer skinned” patients, with the most common location on the lower extremities for women and the trunk for men.^{4,5} American Indian and Alaska Native women had higher rates of superficial spreading melanoma while men had higher rates of nodular melanoma, similar to a report of primarily “fairer skinned” patients.⁴ After accounting for confounders, female sex confers a survival advantage over male sex in studies of all races, and this remained true in American Indian and Alaska Native patients.^{4,5}

Limitations include lack of disease-specific survival, a substantial portion of unspecified histologic subtypes, lack of information regarding specific medical conditions, and potential inaccuracies of reported data in a large database. In addition, American Indian and Alaska Natives are grouped into one category and therefore could not be analyzed separately. Despite limited numbers for each sex, sample size was maximized with the use of a large national cancer registry.

Multiple initiatives have been implemented to address health disparities in American Indian and Alaska Natives, including the Spirit of Eagles, a National Cancer Institute program facilitating culturally relevant cancer education, screening, treatment, and research among American Indian and Alaska Natives. Several academic institutions

have also developed community outreach programs for cancer prevention and management with tribal partners. These partnerships are critical to future community-based education endeavors, exploration of the etiology of sex-related survival disparities, and improving outcomes for the American Indian and Alaska Native population.

References

1. Indian Health Service. *Fact Sheet—Indian Health Disparities*. 2019. Available at: <https://www.ihs.gov/newsroom/factsheets/disparities/>. Accessed October 30, 2022
2. Nguyen B, Bray FN. Access to dermatologic care in Indigenous American communities. *J Am Acad Dermatol* 2022;87:904–6.
3. Wu XC, Eide MJ, King J, Saraiya M, et al. Racial and ethnic variations in incidence and survival of cutaneous melanoma in the United States, 1999–2006. *J Am Acad Dermatol*. 2011;65:S26.e1–S26.e13.
4. Behbahani S, Maddukuri S, Cadwell JB, Lambert WC, Schwartz RA. Gender differences in cutaneous melanoma: demographics, prognostic factors, and survival outcomes. *Dermatol Ther* 2020; 33:e14131.
5. Hieken TJ, Glasgow AE, Enninga EAL, Kottschade LA, et al. Sex-based differences in melanoma survival in a contemporary patient cohort. *J Women's Health* 2020;29:1160–7.

Jennifer M. Fernandez, MD, RD*
Elizabeth M. Mata, MSN, BS†

Jennifer Erdrich, MD, MPH‡
Mohammad Fazel, MD, PharmD§

*Department of Dermatology, University of Nebraska
Medical Center, Omaha, Nebraska
†University of Arizona, College of Medicine—Tucson, Tucson,
Arizona

‡Department of Surgery, University of Arizona, College of
Medicine—Tucson, Tucson, Arizona

§Division of Dermatology, Department of Medicine,
University of Arizona, College of Medicine—Tucson, Tucson,
Arizona

The authors have indicated no significant interest with commercial supporters.

Data availability statement: The National Cancer Database is jointly sponsored by the American College of Surgeons and the American Cancer Society. Data are available to institutions that contribute to the database and are accredited by the Commission on Cancer.

Ethics approval statement: The Office of Regulatory Affairs at University of Nebraska Medical Center determined this project does not constitute human subject research as defined at 45CFR46.102. Therefore, it is not subject to federal regulations and does not require institutional review board review.

Do Indoor LED Grow Lights Emit Sufficient UV Irradiation to Pose an Increased Skin Cancer Risk?

Grow lights are lamps or bulbs designed to replicate the ultraviolet A (UVA) and ultraviolet B (UVB) spectrum of the sun for indoor plant growth. These lights are being marketed online as desk lights for plant owners and have contributed to the expansion of the indoor vertical farming industry (Amazon, October 2021; CBS News, May 2022). Manufacturers claim grow lights do not pose a risk of skin damage without independent verification (Marketing Sansi, July 2021). However, studies have shown that long-term exposure to UV radiation from indoor artificial light sources can lead to cumulative skin damage.^{1,2} The inclusion of UVB in grow lights is especially concerning. UVB is estimated to cause more than 90% of all cutaneous squamous cell carcinomas and can cause measurable DNA damaging events in doses as low as 4 to 50 mJ/cm².³ This study aimed to assess the safety profile of grow lights by determining whether the UV radiation emitted by these lights can reach the cumulative thresholds reported to cause DNA damage to the skin.^{3,4}

Methods

The top 10 LED grow lights (5 lamps and 5 bulbs, ranked highest in popularity) with unique manufacturers were selected from Amazon's bestsellers list (October 2021). An IL700 research radiometer (International Light Technologies Inc., Peabody, MA) was used to measure the UVA and UVB irradiances at 4 distances: 0, 10, 20, and 30 cm from the grow light's center of illumination. These distances are

commonly used for plant growth and are probable distances at which plant owners would place grow lights from their desks (Urban Leaf, April 2021). The measurements taken at these distances also enabled the determination of the relationship between irradiance, and therefore skin cancer risk, with distance from the grow light. In addition, this measurement schema allows confirmation that the experimental setup adheres to the inverse square law, which describes the relationship between irradiance from a point source and distance. The radiometer's accuracy was validated by comparing its measurements of a clinical UVA and UVB phototherapy unit (National Biological Houva II) with established values of 7,000 to 15,000 $\mu\text{W}/\text{cm}^2$ and 400 $\mu\text{W}/\text{cm}^2$ for its narrow spectrum UVA and UVB bulbs, respectively.^{3,4} The median UVA and UVB irradiance of each grow light was used to determine the exposure times to achieve cumulative UV thresholds that have been previously reported to result in DNA damage in irradiated keratinocytes, which is 60 J/cm² (60,000,000 $\mu\text{W} \times \text{s}/\text{cm}^2$) for UVA and 4 mJ/cm² (4,000 $\mu\text{W} \times \text{s}/\text{cm}^2$) for UVB.^{3,4}

Results

The radiometer's accuracy was confirmed by the median readings from the phototherapy unit, which matched the known ranges for its UVA (7,050 $\mu\text{W}/\text{cm}^2$) and UVB (353 $\mu\text{W}/\text{cm}^2$) bulbs. Furthermore, the median UVA and UVB irradiances of the grow lights were significantly lower than